

Curriculum vitae

Sacha Cardonna

Ph.D. candidate in Mathematics

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1. Profile

1.1. Current appointment

Ph.D. candidate in Mathematics

10/2023 - Present

Institut Montpellierain Alexander Grothendieck – Montpellier, France

Title. High-order numerical methods for shallow-water flows on unstructured meshes: moving boundaries and wave-structure interactions.

Advisors. François Vilar & Fabien Marche.

Funding. French research ministry fellowship, ranked 1st at I2S Doctoral School admission exam.

1.2. Research interests

- **Modeling.** Conservation laws, nonlinear hyperbolic systems, dispersive and elliptic equations, coupled and multiphysics models, free-surface flows, moving domains and fluid–structure coupling...
- **Numerical analysis.** High-order finite element and discontinuous Galerkin methods, subcell schemes, hybrid high-order discretizations, well-balanced methods, entropy-preserving and structure-preserving schemes, Arbitrary Lagrangian–Eulerian approaches, methods on manifolds, variational physics-informed neural methods...
- **Scientific computing.** Object-oriented and generic programming, CPU–GPU co-processing and parallel computing...
- **Applications.** Fluid mechanics, water waves and wave–structure interactions, ideal magnetohydrodynamics and Euler–Heat flows, conservation laws on surfaces...

2. Education and qualifications

2.1. Academic education

- **Graduate studies in Mathematics** 09/2020 - 06/2023
Faculty of Sciences – Montpellier, France
Master’s Degree in Theoretical and Numerical Analysis of PDEs (*highest honours*).
Master’s Degree in Fundamental Mathematics.
Rank. Valedictorian (equiv. GPA: 4.0).
- **Undergraduate studies in Mathematics & Physics** 09/2017 - 05/2020
Faculty of Sciences – Montpellier, France
Bachelor’s Degree in Pure and Applied Mathematics (*first-class honours*).
Associate’s Degree in Mathematics & Physics (*first-class honours*).
Rank. Consistently ranked in the top 10% best students (equiv. GPA: 3.7).

3. Research outputs

3.1. Publications

- **Modeling and numerical study of wave-structure interactions in shallow-water regime** 2026
S. Cardonna, D. Lannes, F. Marche & F. Vilar – In preparation
- **A high-order robust subcell monolithic DG/FV formulation for nonlinear shallow-water equations on unstructured grids** 2025
S. Cardonna, F. Marche & F. Vilar – Submitted
- **Local subcell monolithic DG/FV methods for nonlinear shallow-water models with source terms** 2025
S. Cardonna, A. Haidar, F. Marche & F. Vilar – Submitted

4. Professional experience

4.1. Appointments

- ▶ **Ph.D. candidate in Mathematics** 10/2023 - Present
Institut Montpelliérain Alexander Grothendieck – Montpellier, France
- ▶ **Teaching assistant, Department of Mathematics** 10/2023 - Present
Faculty of Sciences & Engineering School Polytech – Montpellier, France
- ▶ **CNRS research intern** 03 - 07/2023
Institut Montpelliérain Alexander Grothendieck – Montpellier, France
- ▶ **Private tutor in Sciences** 07/2017 - 05/2023
Face-to-face and remote courses – Salon-de-Provence & Montpellier, France

4.2. Teaching

- 2025–2026** Faculty of Sciences – Montpellier, France
- ▶ **Numerical analysis of partial differential equations** (1st year *M.Sc.*). Practical classes. *Topics*: finite difference methods, analytical resolution of conservation laws, finite volume methods.
 - ▶ **Linear algebra for sciences and engineering** (1st year *B.Sc.*). Tutorial classes and exam marking. *Topics*: matrices, vector spaces, diagonalization.
 - ▶ **Probabilities for life sciences** (1st year *B.Sc.*). Tutorial classes and exam marking. *Topics*: set theory, discrete probability models, numerical approximations.
- 2024–2025** Faculty of Sciences – Montpellier, France
- ▶ **Numerical analysis of differential equations** (3rd year *B.Sc.*). Tutorial classes, coding, and exam marking. *Topics*: Euler, Runge–Kutta and finite difference schemes; ordinary and partial differential equations.
 - ▶ **Convex optimization** (3rd year *B.Sc.*). Tutorial classes, coding, and exam marking. *Topics*: unconstrained and constrained extrema, applied mathematics, machine learning.
 - ▶ **Real analysis for sciences and engineering** (1st year *B.Sc.*). Tutorial classes, exam writing and marking. *Topics*: logical reasoning, set theory, real-valued functions, integral calculus, first-order differential equations.
- 2023–2024** Engineering School Polytech – Montpellier, France
- ▶ **Differential calculus & multivariable integration** (2nd year *B.Sc.*). Tutorial classes, exam writing and marking. *Topics*: differential calculus, multivariable integration, topology, bilinear algebra.
 - ▶ **Supervised mathematical assignments** (2nd year *B.Sc.*). Supervision of student groups preparing for midterms and final exams. *Topics*: differential calculus, multivariable integration, topology, bilinear algebra.

5. Computational research

5.1. Research software

- ▶ **WaveBox – High-order numerical platform for water-wave equations**
C++, Python | Fabien Marche (lead), Sacha Cardonna, Arnaud Duran, Matthieu Rigal, et al.
Multi-model platform coupling DG, HDG, HHO, and monolithic DG/FV(-ALE) methods on general unstructured meshes for the Saint-Venant, Boussinesq, and Green-Naghdi equations.
Contributions. Developed a new high-order monolithic DG-FV framework with one- and two-dimensional subcell grid subdivisions and a priori stabilization. Developed a multiphysics numerical model for nonlinear wave-structure interactions, coupling hyperbolic flow models with advanced SWIP-DG and HHO discretizations of elliptic problems, as well as a high-order subcell ALE scheme. Developed Python visualization tools and contribute to Git integration and maintenance with Fabien Marche.
- ▶ **Bfree – Structure-preserving finite-volume solver for ideal MHD**
Fortran, Python | W. Boscheri (lead), M. Lupi, E. Bernardelli, L. Gude Vila, S. Cardonna
Research platform for two-dimensional ideal magnetohydrodynamics and Euler-Heat flows, preserving at the discrete level the divergence-free magnetic field and curl-free thermal impulse constraints.
Contributions. Contributed to the design and implementation of the numerical method, in particular its Arbitrary Lagrangian-Eulerian formulation. Developed the nonlinear Newton solver required to resolve the constrained KKT-type system while retaining the underlying discrete differential constraints.
- ▶ **ManicoreFV – Discontinuous Galerkin methods for conservation laws on curved surfaces**
C++, Python | Marien Hanot, Sacha Cardonna
Extension of Marien Hanot’s Manicore framework for conservation laws on manifolds, using local charts, parametrizations, and general Riemannian metrics.
Contributions. Designed and implemented the high-order DG method within the existing geometric framework, including both the spatial discretization and explicit Runge-Kutta time integration with CFL-based time-step

selection. Developed bound-preserving limiting for discontinuous solutions while maintaining local cell averages and global mass conservation.

► **ShoreVPINN – Variational physics-informed neural solver for shallow-water flows**

Python (PyTorch) | Ali Haidar, Sacha Cardonna

Ongoing development of a variational physics-informed neural method for the one-dimensional nonlinear Saint-Venant equations with topography. Local polynomial test functions enforce the equations in weak form over each spatial cell.

Contributions. Joint development with Ali Haidar of a structure-preserving formulation designed to combine lake-at-rest well-balancing, efficient shock capturing, positivity of the water depth, and robustness at moving wet-dry interfaces.

► **DG4SCL – Compact and student-friendly DG code for one-dimensional scalar conservation laws**

C++, Python | Sacha Cardonna

Pedagogical implementation emphasizing a compact, transparent structure for students learning Discontinuous Galerkin methods.

Contributions. Developed the complete code from scratch, with input from Ali Haidar and François Vilar.

6. Recognition and scholarly communication

6.1. Grants and awards

► **Support grants from the I2S Doctoral School**

2023 - 2026

Selected through school-wide competitive calls open to doctoral candidates across all disciplines: **750€** for computer equipment, **750€** for travel to ICOSAHOM, and **300€** for the travel of a Ph.D. defense committee member.

► **I2S Doctoral School scholarship competition**

2023

Highly selective admission examination for a French Ministry of Research Ph.D. scholarship; ranked 1st among all candidates in mathematics. The scholarship provides **125,500€** over three years.

6.2. Conferences, seminars, and workshops

2026

► **SUN HYPE 2026 – Coding and modeling week on and beyond hyperbolic equations**

Technical University of Crete – Chania, Greece

Project. Preserving differential constraints in structure-preserving schemes for ideal magnetohydrodynamics with ALE mesh motion.

► **SHARK-FV – Sharing Higher order Advanced Research Known-how on Finite Volume**

Minho, Portugal

Talk. Subcell monolithic DG/FV-HHO-SSP-RK scheme for a new wave-structure interaction model.

► **Applied Maths Colloquium – Seminar of the Applied Mathematics group at UW**

University of Waterloo – Waterloo, Canada

Seminar. A high-order DG/FV convex property preserving scheme for hyperbolic systems with applications to shallow water flows and fluid-structure interaction.

► **CJC-MA – Congrès des Jeunes Chercheur.e.s en Mathématiques Appliquées**

École nationale des ponts et chaussées – Champs-sur-Marne, France

Talk. Modeling and numerical simulation of floating structures in shallow-water flows.

► **Métiers des mathématiques – Conference for Bachelor's and Master's students**

Montpellier, France

Talk. Modèles et méthodes numériques pour les interactions entre vagues et structures flottantes.

2025

► **ICOSAHOM – International Congress on Spectral and High Order Methods**

McGill University – Montréal, Canada

Talk. Local subcell monolithic DG/FV scheme for NSW equations with source terms on unstructured grids.

► **MIPS Colloquia – Semaine du pôle Mathématiques, Informatique, Physique, Systèmes**

Montpellier, France

Talk. Designing numerical methods for free-surface flows towards reliable wave-structure interactions.

► **CIMAV – Congrès Interdisciplinaire sur les Modèles Avancés de Vagues**

Aussois, France

Talk. A high-order robust DG/FV scheme for nonlinear shallow water equations with source terms on unstructured grids.

2023

- **Introduction to newcomers – Séminaire des Doctorants**
Montpellier, France
Seminar. Modeling, solving & implementing PDEs from wave-structure interactions.

7. Academic service and outreach

7.1. Peer reviewing

- **Journal of Computational Physics (J. Comp. Phys).** Ad hoc reviewer.

7.2. Institutional responsibilities

- **Member of the Sustainable Development Committee at the IMAG laboratory** 2024 - Present
Representing Ph.D. students in meetings and decision-making processes, contributing to the annual GHG (greenhouse gas) inventory with a homemade carbon footprint assessment tool and proposing strategies to improve the lab's environmental impact.
- **Student Representative of Master's Degree in Applied Mathematics** 2023 - 2024
Representing graduate students in various reunions of development council, in order to convey student comments and improve the formation.

7.3. Scientific outreach

2025

- **Jury member for TFFJM² national mathematics competition** – Evaluated high-school teams presenting open-ended research problems; led critical discussions and assessed mathematical reasoning, rigor, collaboration, and scientific communication.

2024

- **Mastering games with Mathematics (MathC2+ program)** – Designed and delivered activities connecting Tic-Tac-Toe to game theory, Mastermind to Knuth's algorithm, and Spot It! to combinatorial designs and projective geometry; developed a program generating personalized Spot It! cards.
- **MATh.en.JEANS congress** – Introduced high-school students to collaborative mathematical research and scientific presentation through a national network of researcher-led workshops.
- **Mathematics & Modeling (MathC2+ program)** – Created hands-on activities combining free-fall modeling, numerical root finding, physical optimization with mini catapults, and a Python solver for quadratic equations.

2023

- **Introduction to Applied Mathematics (MathC2+ program)** – Connected motion modeling and problem solving through a train-collision problem solved analytically and numerically with the bisection method.
- **Fête de la Science** – Presented mathematical careers, doctoral research, and academic pathways to high-school students at Peytavin and Saint-Pierre-Saint-Paul schools.

8. Research training and development

8.1. Research internships

- **Local monolithic DG/FV subcell scheme on nonlinear shallow-water equations** 03 - 07/2023
Building and implementing a new strategy for stabilizing discontinuous Galerkin numerical methods using a Finite-Volume subcells type approach for the nonlinear shallow-water equations.
Advisors. François Vilar & Fabien Marche.
- **Asymptotic analysis of PDEs sequences and homogenization theory** 02 - 05/2022
Analysis of homogenization problems for PDEs on variable domains, with an application to ice-fog formation.
Advisor. Michel Bellieud.
- **From differential geometry to mathematical billiards** 03 - 05/2021
Studying one of the simplest dynamical system, the mathematical billiard where we characterize the periodic trajectories by their initial angle of shot.
Advisor. Daniel Massart.
- **Proof of Dirichlet Prime Number theorem** 01 - 04/2020
Demonstrating that, for $a, b \in \mathbb{N}^*$, such that $\gcd(a, b) = 1$, the arithmetic progression $\{an + b\}_{n \in \mathbb{N}}$ contains an infinity of prime numbers. Such a proof needs various theories, like complex analysis or group theory.
Advisor. Sylvain Brochard.

8.2. Professional development

2024

- **Advanced Programming for Scientific Computing** – Advanced C++, memory management, object-oriented programming, scientific data processing, and collaborative development with version control.
- **Creative Pedagogy (Levels 1 & 2)** – Collaborative learning, interactive teaching, adaptation to diverse learning contexts, and course design.
- **Management & Leadership** – Team management, professional communication, decision-making, and leadership posture.

2023

- **Scientific Writing & Publishing** – Structure of scientific articles, peer review, publication standards, and research integrity.
- **Artificial Intelligence and Ethics** – Privacy, bias, responsible use of AI, and ethical decision-making.
- **Research Ethics** – Scientific integrity, conflicts of interest, institutional responsibilities, and contemporary ethical issues.

9. Skills and additional background

9.1. Technical skills

Programming	C/C++, Python, Fortran.
Mathematics softwares	FreeFEM++, Matlab, Scilab, gnuplot, Maple, Mathematica.
Markup languages	HTML, CSS, PHP.
Typesetting systems	L ^A T _E X, Beamer, Microsoft Office (<i>Word & PowerPoint</i>).
Operating systems	Linux (<i>Kali & Debian</i>), Windows, macOS.
Creation softwares	Adobe Creative Cloud, Audacity, Final Cut Pro X.

9.2. Selected academic projects

2022

- **Hybrid High-Order method on Leray-Lions operators** – Discrete functional analysis and non-conforming finite-element approximation of nonlinear elliptic problems.
- **Müller’s SPH C++ implementation for fluid dynamics** – Implementation of a Smoothed Particle Hydrodynamics solver.
- **Some proofs in measure theory** – Differentiation of Radon measures; Besicovitch, Vitali, Tietze, and Lusin theorems.
- **Finite-element simulation with FreeFEM++** – Dirichlet problem with mixed boundary conditions.

2021

- **Machine-learning code for database analysis** – Data analysis and regression methods implemented in Python.

2018

- **Numerical interpolation and its limits** – Polynomial interpolation and Runge’s phenomenon.

9.3. Selected coursework

- **Fundamental courses** – Theoretical Analysis of PDEs, Functional Analysis & Distribution Theory, Differential Geometry, Measure and Integration Theory, Complex Analysis, Topology of Metric Spaces, Galois Theory, Category Theory, Ring & Group Theory, Differential Equations & Calculus, Probability Theory, Euclidean Geometry, Linear & Bilinear Algebra, Real Analysis
- **Applied and specialized courses** – Numerical Analysis of PDEs, Numerical Modeling, Homogenization for Navier-Stokes, Scientific Computing, Machine Learning & Convex Optimization, *A Posteriori* Estimates & Mesh Adaptation, Fourier Transform & Convolution for Inverse Problems, Deterministic & Stochastic Modeling
- **Physics courses** – Solid & Fluid Mechanics, Electromagnetism, Electrostatics & Magnetostatics, Thermodynamics, Wave & Geometrical Optics, Electrohydrodynamics, Experimental Physics

9.4. Languages

- **French:** native (TEF C2)
- **English:** fluent (IELTS 7.5)
- **Spanish:** intermediate ($\simeq$ B2)
- **Japanese:** beginner ($\simeq$ A1)